```
#function assignment
#write function is prime(n) takes that function number and return true if its p
def is_prime(n):
  if n<2:
    return False
  for i in range(2,n):
     if n%i==0:
        return False
  return True

print("Prime number 1 to 50")
for num in range(1,51):
    if is_prime(num):
        print(num)
```

```
Prime number 1 to 50
2
3
5
7
11
13
17
19
23
29
31
37
41
43
47
```

```
# Question 2 function argument
#Write a function calculate_bill(item, price, quanti
def calculate_bill(item, price, quantity=1, discount
    total = price * quantity
    total = total - (total * discount / 100)
    print(item, "=", total)


# 1. Positional arguments
calculate_bill("laptop", 45000, 1, 5)

# 2. Keyword arguments (different order)
calculate_bill(discount=20, item="Book", price=100,

# 3. Default quantity and discount
calculate_bill("Pen", 5)
```

```
laptop = 42750.0
Book = 160.0
Pen = 5.0
```

```
#Question 3 — Lambda Functions Write a lambda function that takes a number and
square=lambda x:x**2
numbers=[12,6,13,24,55]
for num in numbers:
    print(square(num))
```

```
144
36
169
576
3025
```

```
#Recursion Write a recursive function fibonacci(n) that returns the nth Fibonac
def fibonacci(n):
    # Base case
    if n == 0:
        return 0
    if n == 1:
        return 1

    # Recursive case
    return fibonacci(n - 1) + fibonacci(n - 2)


for i in range(10):
    print(fibonacci(i))
```

```
0
1
1
2
3
5
8
13
21
34
```

```
#Question 5 — Nested Functions & Returning Functions
def make_multiplier(factor):

    def multiply(x):
        return x * factor

    return multiply
```

```
double = make_multiplier(2)
triple = make_multiplier(3)

print(double(5))
print(double(10))

print(triple(5))
print(triple(10))
```

```
10
20
15
30
```